

- Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
2) Use of design data hand book is permitted.

Unit – I

Marks BT* CO* PO*

1. a) Define codes & Standards with an examples and also explain the general procedure in machine design with block diagram.
- b) A steel member is loaded as shown in Fig. Q1 (b). Determine the magnitude of
 - i) Maximum normal stress
 - ii) Minimum normal stress
 - iii) Maximum shear stress

10 L*1 01 01

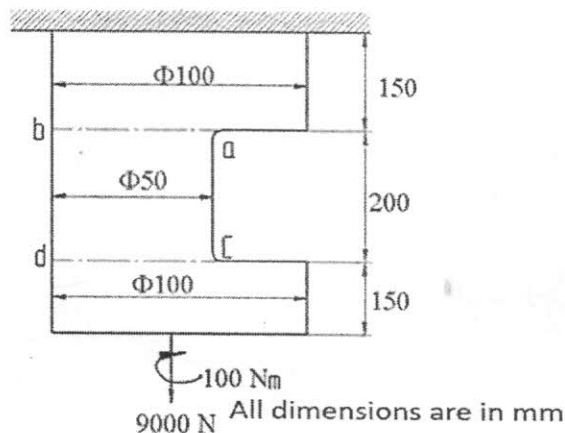


Fig. Q1 (b)

10 L3 01 01

2. a) Define stress concentration? How can it be reduced with suitable design practice?
- b) Determine the maximum normal stress and maximum shear stress at section A-A for crank shown in Fig.Q2 (b). When load of 10kN is assumed to be concentrated at the center of crank pin.

04 L1 01 01

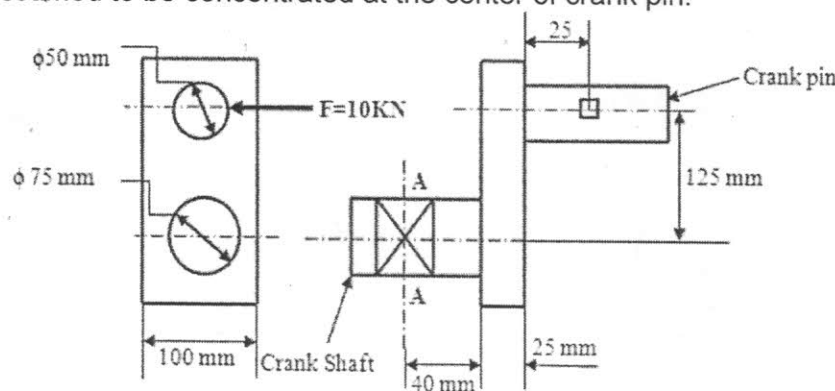


Fig.Q2 (b)

08 L3 01 01

- c) A cylinder shaft made up of steel having $\sigma_y = 700 \text{ N/mm}^2$ is subjected to static load consisting of bending moment 10 kNm and torsional moment 30kNm. Determine the shaft diameter based on maximum normal stress theory. Assume FOS=2.

08 L2 01 01

Unit – II

3. a) Define fatigue load. Illustrate Goodman design equation when a machine member is subjected to fatigue loading.
- b) A SAE 1025 cold drawn water quenched steel rod [$\sigma_u=620.8\text{MPa}$, $\sigma_y=400.1\text{MPa}$, $\sigma_{en}=345.2\text{MPa}$] of circular cross-section shown in Fig.Q3 (b). Is subjected to load that varying from $-P$ to $3P$. Determine the value of P , use factor of safety of 3.

08 L2 02 01

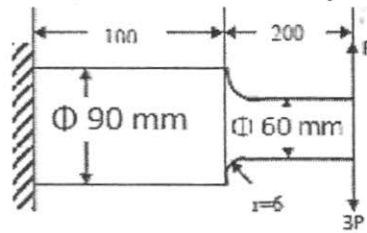


Fig.Q3 (b)

12 L3 02 02

4. a) With the help of S-N diagram explain briefly
- Low cycle fatigue
 - High cycle fatigue
 - Endurance limit
- b) A cold rolled steel rod of circular cross-section is subjected to a variable bending moment of 565N-m to 1130N-m as the axial load varies from 4500N to 13500N. The maximum bending moment occurs at the same instance when the axial load is maximum. Determine the diameter of rod for $n=2$, $\sigma_u=550\text{MPa}$, $\sigma_y=470\text{MPa}$.

06 L1 02 01

14 L3 02 02

Unit – III

5. A commercial shaft is supported by two bearings 1500mm apart. A keyed gear 20° involutes and 175mm in diameter is located 400mm to left of the right bearing and is driven by a gear directly behind it. A 600mm diameter pulley is keyed to the shaft 600mm to the right of the left bearing and drives a pulley with a horizontal belt directly behind it. The tension ratio of belt is 3 to 1 with a slack side on the top. The drive transmits 45 kW at 330 rpm. Take $G=80\text{GPa}$, and $C_m=C_t=1.5$. Calculate the shaft size. Assume direction of viewing from right.
6. A hollow shaft is supported by two bearings placed 1.2 m apart. A 600 mm diameter pulley is mounted at a distance of 200 mm to the right of left hand bearing and this drives a pulley directly below it with the help of a belt having maximum tension of 2 kN. Another pulley of 400 mm diameter is placed 200 mm to the left of right bearing and is driven with the help of a motor and belt which is placed at 45° to vertical and downward towards the observer. The angle of contact for both pulleys is 180° and coefficient of friction is 0.25. Determine the suitable diameter for hollow shaft having diameters ratio of 0.5. The allowable tensile and shear stresses are 63 MPa and 42 MPa respectively. Take $C_m = 2.5$ and $C_t=2$. Assume direction of viewing from left.

20 L3 03 02

20 L3 03 02

Unit – IV

7. a) A Bracket is fixed on a steel column as shown in Fig.Q7 (a) for supporting a travelling crane in a workshop. The maximum load that comes on the bracket is 12kN, acting vertically at a distance of 400mm from the face of the column.

10 L3 04 02

The vertical face of the bracket is secured to the column by four bolts in two rows [two in each row] at a distance of 50mm from the lower edge of the bracket. Determine the size of the bolts if the permissible value of the tensile stress for the bolt material is 84 N/mm^2 . Also find the cross section of the arm of the bracket which is rectangular. Assume the depth of the arm of the bracket as 250mm.

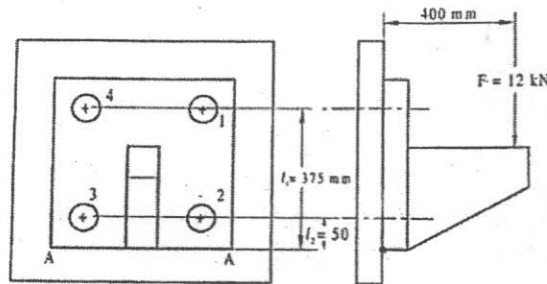


Fig.Q7(a)

- b) An M10 steel bolt of 125 mm long is subjected to an impact load. The kinetic energy absorbed by the bolt is 2.5 J, take $E=206.8\text{ GPa}$. Determine,
- Stress in the shank of the bolt if there is no threaded portion between the nut and the bolt head.
 - Stress in the shank if the area of the shank is reduced to that of the root area of the thread or the entire length of bolt is treaded.

10 L3 04 02

8. a) Explain about the stresses induced in threaded fasteners and describe the design procedure for eccentrically loaded bolted joint.
- b) Design a rigid flange coupling to transmit 18 kW at 1440 rpm. The allowable shear stress for CI flange is 4 MPa. The shafts, keys and bolts are made of annealed steel having allowable shear stress of 93 MPa. Allowable crushing stress for key = 186 MPa.

10 L2 04 01

10 L3 04 02

Unit – V

9. a) Differentiate between Lap joint and butt joint of rivets and describe tearing and shearing of riveted joints with neat sketches.
- b) Design a single riveted single cover butt joint to connect two plates 30 mm thick. The yield stress for plate and rivet materials is 324 MPa. Use factor of safety of 2.5 for tension, 4 for shear and 2 for crushing.
10. a) What are welded joint? Explain with sketches butt joint and lap joint.
- b) Determine the size of weld for a bracket welded as shown in Fig. Q10 (b). The allowable stress in the weld is 75 MPa.

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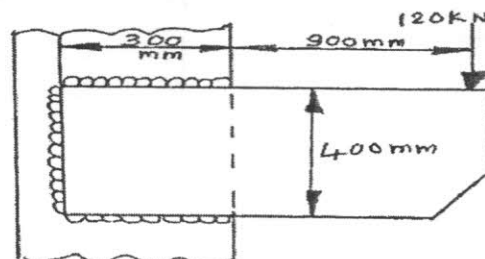


Fig. Q10 (b)

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